

Summary

Over the past ten years, I have been working as an assistant professor in physics at the National Institute of Technology Karnataka (NITK) in India. While here I have taught introductory level physics courses and graduate level courses in mathematical physics, statistical mechanics, nuclear physics and general relativity. I have played a crucial role in implementing a more student friendly approach in our teaching practices.

In the following now I outline my approach towards teaching, after which I detail the teaching activities I have engaged in over the course of my career.

Teaching Philosophy

The best teachers are not those who know how to stand in front of a class and deliver hour long lectures or solve difficult problems on the blackboard. To be a good teacher means to understand your students, to sense their needs and to convey the joy and excitement of learning which you yourself experienced when first learning the subject. Those who have never felt the high of that moment when something first "clicks", when your perspective on some part of existence is forever changed in an instant, cannot in turn be good teachers. For they cannot convey to the student a feeling which is alien to themselves. And it is the experience of this feeling, of the sheer joy of discovery, which makes learning and teaching a truly rewarding experience.

The following is a summary of my teaching philosophy and what I believe it takes to be an effective instructor.

1. **To uplift the weakest:** Teaching should be designed to uplift the weakest of students, rather than to strengthen the skills of those who are already capable. In any type of class-based, collective mode of instruction, there will always be a spread in the capabilities of the students. Nothing pleases a teacher's ego more than to interact with the smartest kids in class, the ones who are most willing, eager and capable of learning. The opinions of the best students are often used as a barometer to measure the effectiveness of a teacher. However, doing so is a disservice of our position as teachers. The "smart" kids will manage some way or the other. The curriculum and teaching methods should be targeted to the weakest students. It is those who need the greatest help. If we let them fall behind we betray our fundamental duty as teachers.
2. **To encourage the exceptionally talented:** Adhering strictly to the first point can sometimes have a negative effect on the morale of those students who are naturally talented. While ensuring equity and fairness in education is a worthwhile goal intended for the benefit of the many, such an approach will necessarily lead to dissatisfaction and disenchantment with the education system in those who are extremely talented. The peculiar talents of exceptional students cannot be neglected. Their talent must be recognized, encouraged and they must be provided

with opportunities to learn at a rate more suited to them.

3. **To ensure equitable treatment:** In any modern society, students in any institution come from a diversity of social, economic, linguistic and religious backgrounds. Unfortunately technological limitations do not yet allow us to overcome linguistic barriers. There must be a preferred language for instruction. What can be done, is to ensure that every student is given sufficient and adequate opportunity to improve their skills in the language used for instruction. Further, instructors must also be sensitive to the varying religious beliefs and backgrounds of their students. Humour is an essential ingredient for maintaining students' attention in class. While an instructor cannot submit their freedom of expression to the altar of political correctness, one should be careful not to indulge in jokes that unnecessarily denigrate any community or gender. Knowing how to provoke students to think and question their own beliefs without offending them or making them feel targeted, is an important skill one must have to be a good teacher.
4. **To make learning fun:** "Fun" is often thought of as being inherently contradictory to being a productive worker or to being a good student. This is based on an incorrect assumption about human nature. I believe that we are at our most productive when we enjoy the task entrusted to us. This is not to say that the learning process is or can always consist of pure joy. It will necessarily be, in turn, tedious, slow-moving and frustrating. However, at the end of all the tedium and hard work there must lie some promise of excitement and a sense of discovery. It is the teacher's task to generate and to maintain such a hope in students', whether by various means which could include inspirational stories, quotes, videos, hands-on activities, in class demonstrations, class trips to external facilities or inviting external speakers.